

Solution Architecture

Date	26 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	5 Marks

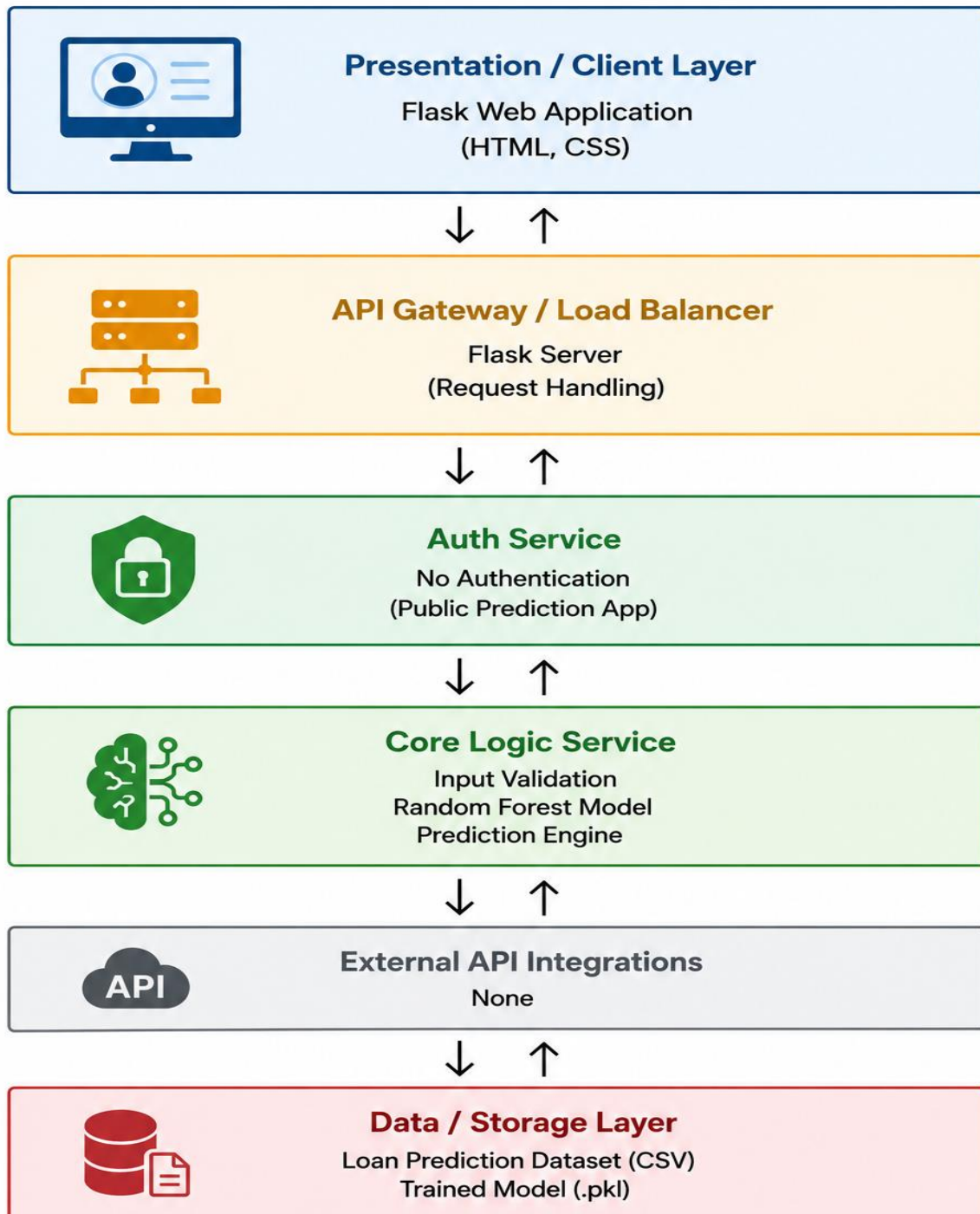
Solution Architecture Diagram:

The Solution Architecture illustrates the overall structure of the Smart Lender – Loan Approval Prediction System. It shows how the presentation layer, application layer, and data layer work together to process user input, perform machine learning predictions, and display the loan approval result efficiently.

The Smart Lender – Loan Approval Prediction System follows a simple three-layer architecture consisting of the presentation layer, application layer, and data layer. The presentation layer allows users to interact with the system through a web interface developed using HTML, CSS, and Flask. The application layer processes user requests, validates the entered information, and uses a trained Random Forest machine learning model to predict loan approval. The data layer contains the loan prediction dataset used during model training and the trained Pickle model used for inference. This architecture ensures smooth communication between system components while providing accurate, efficient, and reliable loan eligibility predictions. The modular design also allows future enhancements and simplifies maintenance and deployment.

Solution Architecture Diagram

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for entering loan details and displaying prediction results.</i>	<i>HTML5, CSS3, Flask</i>
<i>API Gateway</i>	<i>Receives user requests and forwards them to the prediction logic.</i>	<i>Flask</i>
<i>Core Logic Service</i>	<i>Validates user input, loads the trained model, and predicts loan approval.</i>	<i>Python, Scikit-learn, Pickle</i>
<i>Database / Data Layer</i>	<i>Stores the dataset used for training and the trained machine learning model used for prediction.</i>	<i>CSV Dataset, Pickle (.pkl)</i>